

Adding and subtracting vectors

Nose to tail, or by the parallelogram — two pictures of one operation.

LESSON 40–41

2 × 40 MIN

GEOMETRY 8, ТЕМЫ 36–37

STAGE 9 · 13.1–13.2

LESSON CLOCK

6'

14'

12'

4'

4'

Two walks, one displacement	6 min
The triangle and parallelogram rules	14 min
Subtraction	12 min
In coordinates	4 min
Homework	4 min

LEARNING OBJECTIVES

- Add two vectors by the triangle rule and by the parallelogram rule.
- Subtract a vector by adding its opposite.
- Use the relation $AB + BC = AC$.
- Add and subtract in coordinates.

TEXTBOOK

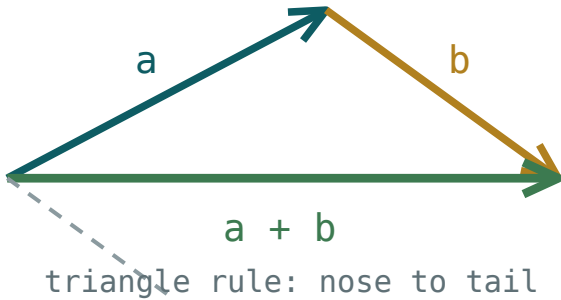
National	Темы 36–37, pp. 81–84
Cambridge	Learner’s Book pp. 284–296
Workbook	Workbook 13.1–13.2

01

Explanation

Adding: nose to tail

Walk 3 km east, then 4 km north. Your displacement is not 7 km — it is 5 km north-east. That is vector addition, and the picture is the whole rule:



The triangle rule: place the second vector at the tip of the first, and the sum runs from the first tail to the last tip.

TRIANGLE RULE

$$AB + BC = AC$$

Whatever route you take, only the start and the end matter.

The **parallelogram rule** gives the same answer with both vectors drawn from a common point: complete the parallelogram, and the sum is the diagonal from that point.

In coordinates, add componentwise:

$$(a_1; a_2) + (b_1; b_2) = (a_1 + b_1; a_2 + b_2)$$

Addition is **commutative** — $a + b = b + a$ — which is exactly why the parallelogram closes.

Subtracting

To subtract, add the opposite:

$$a - b = a + (-b)$$

In coordinates that is simply $(a_1 - b_1; a_2 - b_2)$. In a picture, $a - b$ is the vector **from the tip of b to the tip of a** when both are drawn from the same point — the other diagonal of the parallelogram.

THE RELATION WORTH MEMORISING

$$AB = OB - OA$$

Any vector is the position of its end minus the position of its start. That single line turns most vector problems into subtraction.

LENGTHS DO NOT ADD

$|a + b|$ is **not** $|a| + |b|$ unless the two point the same way. For $(3; 0)$ and $(0; 4)$ the lengths are 3 and 4, but the sum has length 5.

02 Worked examples

EXAMPLE 1 $a = (3; -1)$, $b = (-5; 4)$. Find $a + b$, $a - b$ and $|a + b|$.

1 $a + b = (3 - 5; -1 + 4) = (-2; 3)$

Add componentwise.

2 $a - b = (3 + 5; -1 - 4) = (8; -5)$

Subtract componentwise.

3 $|(-2; 3)| = \sqrt{4 + 9} = \sqrt{13}$

Leave it exact.

ANSWER

$(-2; 3), (8; -5), \sqrt{13}$

EXAMPLE 2 In a parallelogram $ABCD$, express AC and BD in terms of $AB = a$ and $AD = b$.

1 $AC = AB + BC$

Triangle rule, going round through B .

2 $BC = AD = b$

Opposite sides of a parallelogram are equal vectors.

3 $AC = a + b$

One diagonal is the sum.

4 $BD = BA + AD = -a + b$

The other diagonal is the difference.

ANSWER

$AC = a + b, BD = b - a$

EXAMPLE 3 A boat heads north at 8 km/h across a river flowing east at 6 km/h . Find its actual speed.

① $(0; 8) + (6; 0) = (6; 8)$

The two velocities add as vectors.

② $|(6; 8)| = \sqrt{36 + 64}$

③ $= 10\text{ km/h}$

Not 14 — the directions differ.

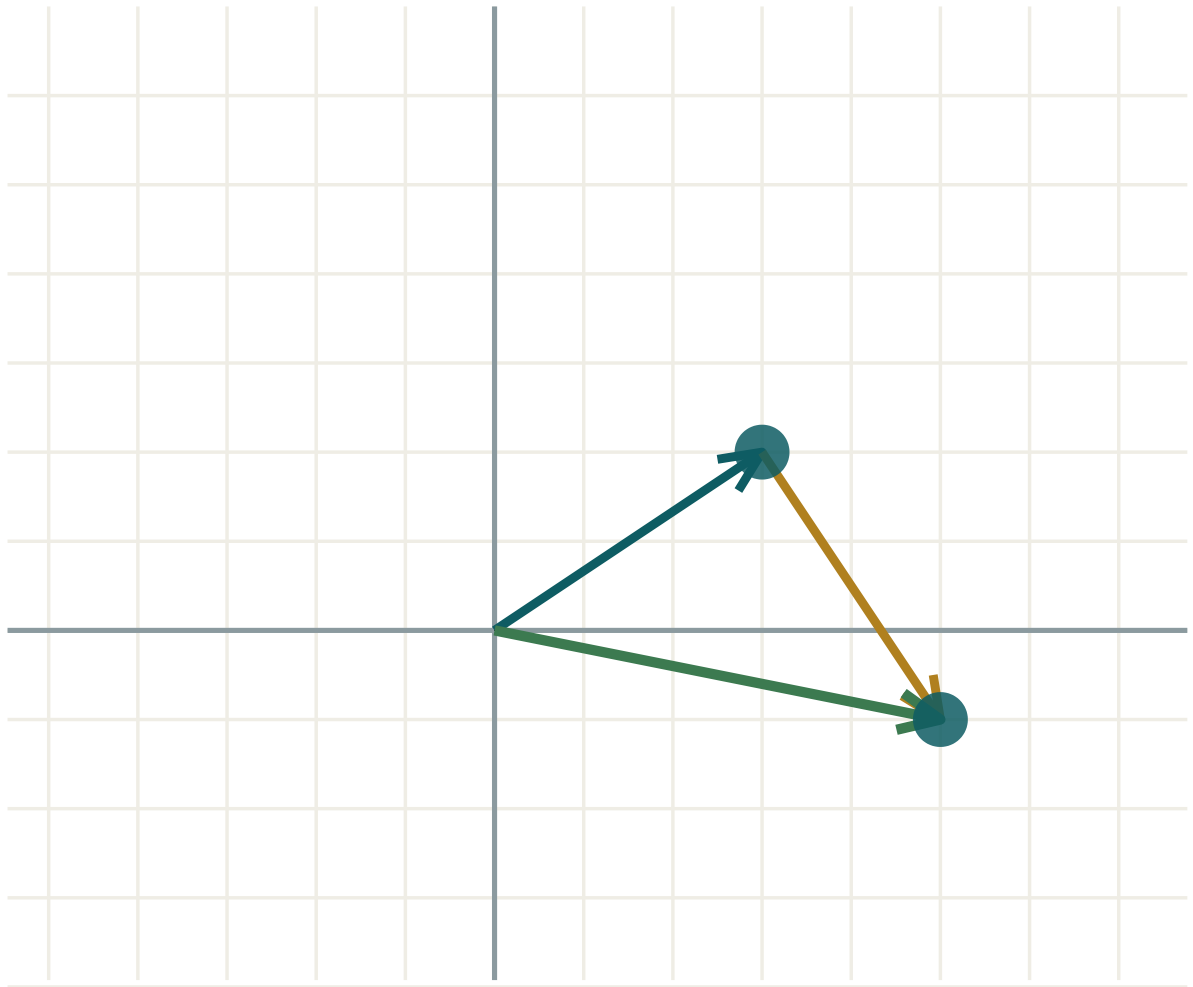
ANSWER

10 km/h

03 Interactive model

Drag the tips of a and b; the green arrow is always their sum, and it closes the triangle.

The triangle rule in action



$a \begin{pmatrix} 3 \\ 2 \end{pmatrix}$

$b \begin{pmatrix} 2 \\ -3 \end{pmatrix}$

$a + b \begin{pmatrix} 5 \\ -1 \end{pmatrix}$

$|a| \ 3.61$

$|b| \ 3.61$

$|a + b| \ 5.1$

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Sum of vectors	Vektorlar yig'indisi	Сумма векторов
Difference of vectors	Vektorlar ayirmasi	Разность векторов
Triangle rule	Uchburchak qoidasi	Правило треугольника
Parallelogram rule	Parallelogramm qoidasi	Правило параллелограмма
Resultant	Natijaviy vektor	Равнодействующий вектор
Nose to tail	Uchi-boshiga	Конец к началу
Commutative	O'rin almashinuvchi	Коммутативный
Associative	Guruhlanuvchi	Ассоциативный
Closed polygon	Yopiq ko'pburchak	Замкнутый многоугольник

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

$(2; 5) + (3; -1)$ is:

$(5; 4)$

$(5; 6)$

$(-1; 6)$

$(6; -5)$

$(4; 1) - (6; 5)$ is:

$(10; 6)$

$(-2; -4)$

$(2; 4)$

$(-2; 4)$

$AB + BC$ equals:

AC

CA

BA

0

If $|a| = 3$ and $|b| = 4$ and they are perpendicular, $|a + b|$ is:

7

5

1

12

AB in terms of position vectors is:

$OA + OB$

$OB - OA$

$OA - OB$

$OA \cdot OB$

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Find $(1; 2) + (3; 4)$.

ANSWER $(4; 6)$

2. Find $(5; 7) - (2; 3)$.

ANSWER $(3; 4)$

3. Find $(0; 0) + (-2; 6)$.

ANSWER $(-2; 6)$

4. Find $(4; -1) + (-4; 1)$.

ANSWER $(0; 0)$

5. Simplify $AB + BC$.

ANSWER AC

6. Find $|(3; 4) + (0; 0)|$.

ANSWER 5

7. Find $(7; 2) - (7; 2)$.

ANSWER $(0; 0)$

Medium · the standard question

7 PROBLEMS

1. $a = (2; -3)$, $b = (-1; 5)$. Find $a + b$ and $a - b$.

ANSWER $(1; 2)$ and $(3; -8)$

2. Find $|(3; 4) + (4; -3)|$.

ANSWER $|(7; 1)| = \sqrt{50} = 5\sqrt{2}$

3. Simplify $AB + BC + CD$.

ANSWER AD

4. Simplify $AB + BA$.

ANSWER 0 — the zero vector

5. $OA = (1; 2)$, $OB = (5; 9)$. Find AB .

ANSWER $(4; 7)$

6. A plane flies north at 200 km/h in a wind blowing east at 150 km/h. Find its actual speed.

ANSWER 250 km/h

7. $a = (6; 8)$. Find a vector b with $a + b = (0; 0)$.

ANSWER $(-6; -8)$

Hard · stretch and reason

7 PROBLEMS

1. In parallelogram $ABCD$, $AB = a$ and $AD = b$. Express DB and CA .

ANSWER $DB = a - b$, $CA = -a - b$

2. Simplify $AB + BC + CD + DA$.

ANSWER 0 — the walk returns to its start

3. $|a| = 5$, $|b| = 12$, and $a \perp b$. Find $|a + b|$ and $|a - b|$.

ANSWER Both 13 — the diagonals of a rectangle are equal.

4. $a = (3; 1)$, $b = (-1; 4)$. Find $|a + b|$ and $|a| + |b|$, and compare.

ANSWER $|(2; 5)| = \sqrt{29} \approx 5.39$ against $\sqrt{10} + \sqrt{17} \approx 7.28$ — the sum is shorter

5. M is the midpoint of AB . Show that $OM = \frac{1}{2}(OA + OB)$.

ANSWER $OM = OA + \frac{1}{2}AB = OA + \frac{1}{2}(OB - OA) = \frac{1}{2}(OA + OB)$

6. Two forces of 10 N and 10 N act at 90° . Find the resultant.

ANSWER $10\sqrt{2} \approx 14.1\text{ N}$ at 45° to each

7. Explain why $|a + b| \leq |a| + |b|$ always.

ANSWER The sum closes a triangle, and in any triangle one side is never longer than the other two together; equality only when the vectors point the same way.

07 Homework

Homework — 6 problems

Geometry 8, Темы 36–37, pp. 81–84. Draw the triangle for every addition.

- Find $(4; -2) + (-7; 5)$ and $(4; -2) - (-7; 5)$.
- Find $|(5; 12) + (0; 1)|$.
- Simplify $AB + BC + CA$.
- $OA = (2; -1)$, $OB = (6; 7)$. Find AB and $|AB|$.

5. *A swimmer heads north at 3 km/h in a current flowing east at 4 km/h. Find the actual speed.*
6. *In parallelogram $ABCD$ with $AB = a$ and $AD = b$, express AC and BD .*